

MIE 402 Lab 1

Dynamic Data Sampling and Frequency Analysis

Fall 2026 | Prof. Ehsan Roohi

In this experiment you will use a microphone and Moku:Go Data Logger to examine how sample rate changes the stored representation of sound. You will compare time histories, frequency spectra, and replayed sound for a nearly tonal source, a harmonic-rich source, and a transient source.

Learning objectives

- Use the Moku:Go Data Logger to acquire and export repeatable voltage measurements.
- Relate signal frequency, sample rate, sample interval, Nyquist frequency, sample count, and record duration.
- Recognize aliasing and other misleading low-rate results in waveforms and spectra.
- Calculate mean, RMS, FFT-bin spacing, and a one-sided amplitude spectrum.
- Explain what can and cannot be concluded from sound replay.

Equipment and supplied files

- Moku:Go used as a data-acquisition device
- Microphone, required adapters/cables, and a computer running the Moku application
- Tuning fork, whistle, and finger snap or GTA-approved sound sources
- MIE402_Lab1_Analyze_and_PlaySound.m for checking exported MAT files

Required recordings

Collect all required cases for at least one second each. Here f is a stated reference frequency for the source; f_s is the recorder sample rate.

Source	How to define f	Required f_s cases	Duration
Tuning fork	Use the frequency printed on the fork.	$100f$, $10f$, $3f$, $1.4f$, f	At least 1 s each
Whistle	Use a high-rate pilot record to identify and state a reference; include relevant harmonics in the sampling decision.	$100f$, $10f$, $3f$, $1.4f$, f	At least 1 s each
Finger snap	A snap is broadband. State the GTA-approved reference frequency or analysis band used to construct the rate matrix.	$100f$, $10f$, $3f$, $1.4f$, f	At least 1 s each

Before recording

1. Match the Moku:Go serial number to your station and open Data Logger.
2. Connect the microphone to the assigned BNC input. Enable only the input channel you use.
3. Confirm that the microphone signal is visible using a high sample rate. If it is flat, check the channel, cable, microphone excitation, source, and input settings.
4. Keep microphone distance, orientation, gain/range, and source conditions as consistent as practical. Record any condition that cannot be held constant.

Data Logger procedure

5. Enter the required sample rate under Acquisition and record the acquisition mode, such as Precision.
6. Choose Immediate or Delayed logging. Set a duration of at least one second.
7. Use a filename that identifies source, reference frequency, sample rate, and trial, for example fork256_fs2560_trial01.
8. Record the sound. Inspect the trace immediately for silence, missed transients, excessive noise, and flat-topped clipping.
9. Repeat until the full three-source by five-rate matrix is complete. Do not overwrite earlier trials.

Export and verify the data

10. Open Moku File Manager, select the recording, choose Convert to MATLAB, select the destination folder, and download the MAT file.
11. Run MIE402_Lab1_Analyze_and_PlaySound.m. Select one MAT file when prompted.
12. Confirm the achieved sample rate from the time-column spacing. Record N, duration, mean, RMS, FFT-bin spacing, and relevant spectral peaks.
13. Save a labeled waveform and spectrum. Repeat the analysis for the required files.
14. Replay the listening copy when supported. Use the achieved sample rate; never use the sample count N as the playback rate.
15. Back up all raw MAT files, the analysis file, exported figures, and acquisition notes before leaving.

Required GTA check before leaving

Show the GTA one valid high-rate tuning-fork waveform and its frequency spectrum. The spectrum must contain a clear peak close to the frequency printed on the fork. Be prepared to identify the achieved sample rate, record duration, and FFT-bin spacing.

Analysis required for the Lab 1 report

- Compare mean and RMS across rates using voltage references. Do not compare RMS in volts with a frequency in hertz.
- Compare the five time-domain waveforms for each sound using consistent axes and sample markers where helpful.
- Compare the spectra, predict aliases, and explain why the physical source did not change when the recorded frequency changed.
- Compare replayed sounds qualitatively. Distinguish acquisition aliasing from playing a correct recording at the wrong rate.
- Explain the difference between a nearly sinusoidal tuning fork, a whistle with harmonics, and a broadband transient snap.

- Recommend a suitable sample rate for each source and justify a practical margin above the highest relevant frequency.

Key equations

- Sample interval: $\Delta t = 1/f_s$
- Nyquist frequency: $f_N = f_s/2$
- Record duration: $T_{\text{record}} = N/f_s$
- FFT-bin spacing: $\Delta f = f_s/N = 1/T_{\text{record}}$
- Mean: $\mu = (1/N) \sum x[n]$
- RMS: $x_{\text{RMS}} = \sqrt{(1/N) \sum x[n]^2}$
- Alias frequency: $f_{\text{alias}} = |f - k f_s|$, choosing integer k so $0 \leq f_{\text{alias}} \leq f_s/2$

Safety and data-quality reminders

- Avoid excessive speaker volume and do not place a loud source directly against the microphone.
- Do not change gain, distance, and sample rate simultaneously unless the change is documented and justified.
- A flat low-rate trace may be an undersampling result, not silence. Verify the setup with a known high-rate record.
- A plausible RMS value does not prove that the frequency content is correct. Inspect both waveform and spectrum.

Submission

Use the current MIE 402 report template and the separate Lab 1 Report Instructions. Submit through Canvas. Canvas controls the due date, file limits, group naming, and any updated requirement.